

ABOUT US

Gravity was founded in South Africa in 1999. Since then Gravity has been a leader in the work at height industry. Assisting in developing a variety of standards, providing excellent training, producing innovative solutions for work sites as well as a range of fall arrest equipment.

Gravity is committed to providing quality services and products with our clients satisfaction being our highest priority.

INDEX

P1. General Site Safety

A. Fall Protection Plan

- Fall Protection Plan 1
- Risk Assessment 2
- Pre-mitigation Risk Rating 3
- Risk-mitigation Hierarchy 4
- Post-mitigation Risk Rating 6

B. Work at Height Basics

- Shock Load 7
- Exclusion Zones and
Barricading 8
- Work Restraint 9
- Rooftop Work 10
- Pitched Roof Work 11
- Fall Arrest 12
- Full Range of PPE 13
 - Helmet 14
 - Harness 15
 - Shock Absorber 17
 - Work Positioning Lanyard 19

INDEX

• - Rope Grab	21
- Descending Devices.....	22
- Slings	23
- Connectors	24
- Work Ropes.....	25
- Retractable Lanyards	26
-Using Retractable.....	27
• Carrying Tools at Height.....	28
• Direct Lifting.....	29
• Pre-use Inspection	30
• Equipment storage	31
• Rope Bagging	32
• Reporting Suspect Equipment	33

D. Work at Height Safety Principles

• Fall Factors.....	34
• Minimum Free Space	35

E. Anchor Points and Lifelines

• Anchor Points.....	36
• Structure Evaluation.....	37
• Temp. Vertical Lifelines	38
• Temp. Horizontal Lifelines	39

G. Emergency Procedures

• Suspension Trauma.....	42
Recovery Position	44

H. Knots

• Klemheist.....	45
• The Way Forward	46

P1. General Site Safety

1. Always wear seat belts when travelling in vehicles.
2. Always use suitable personal protective equipment.
3. Never carry out electrical work on electrical equipment unless competent to do so.
4. Never work under the influence of substances.
5. Never exceed speed limits.
6. Never use a hand held phone whilst driving.
7. Always wear helmets when travelling on motorbike.
8. No passengers on the back of bakkies and trucks.
9. Never undertake any street or underground work activities unless competent to do so.
10. When working in the proximity of power, we always maintain the required safe distance and use the correct insulated equipment



A. Fall Protection Plan

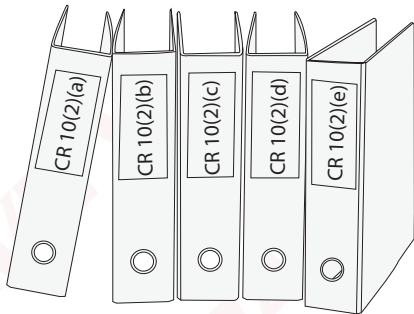
Before any work at height may commence, all the fall risks on site must be addressed and documented in a fall protection plan (FPP).

According to CR10(2) the Fall Protection Plan must include:

- CR10(2)(a) - **Risk Assessment**
- CR10(2)(b) - **Medical Fitness**
- CR10(2)(c) - **Training Programme**
- CR10(2)(d) - **Equipment Management**
- CR10(2)(e) - **Rescue Plan**

A FPP must also be reviewed after any changes on site, or should the work scope change.

Note: All this must be done BEFORE work may start.



1. Risk Assessment

The most important part of the Fall Protection Plan is the **Risk Assessment**.

What to do when approaching a work site

STEP 1 : RISK IDENTIFICATION

The following are factors must be considered when identifying risks:

- **Site requirements**
- **Access method requirements**
- **Required team composition**
- **Fall protection requirements**
- **PPE requirements**
- **Environmental**

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After this, the risks must be communicated to everyone on site by means of:

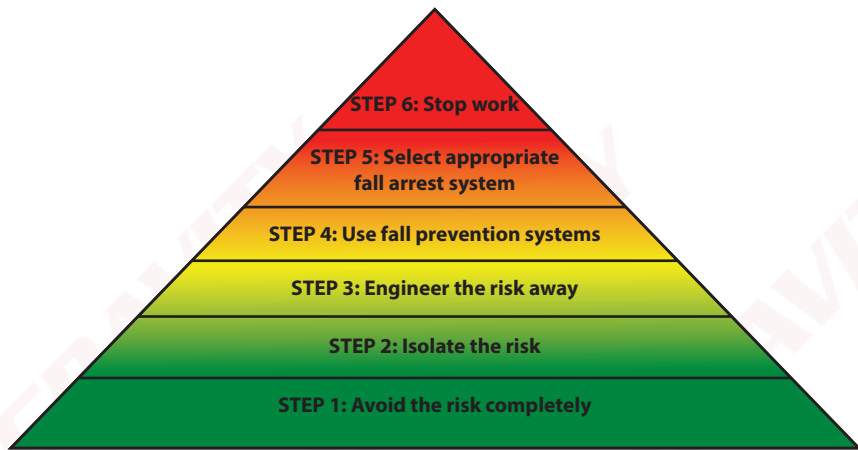
- Site induction
- Toolbox talk

STEP 2 : PRE-MITIGATION RISK RATING

		HAZARD SEVERITY				
		Negligible	Slight	Moderate	High	Very High
LIKELIHOOD OF OCCURRENCE	Very Unlikely	LOW	LOW	LOW	LOW	LOW
	Unlikely	LOW	LOW	LOW	MEDIUM	MEDIUM
	Possible	LOW	LOW	MEDIUM	MEDIUM	HIGH
	Likely	LOW	MEDIUM	MEDIUM	HIGH	HIGH
	Very Likely	MEDIUM	MEDIUM	HIGH	HIGH	HIGH

No work is allowed to commence if there is a "Medium" or "High" risk present on site. Thus, we have to mitigate all identified risks in order for them to receive a "Low" risk rating.

STEP 3 : RISK MITIGATION HIERARCHY



- **Hazardous Environmental Conditions:**

No work-at-height shall be done during any inclement weather conditions. Should the situation change whilst at height, all technicians must safely descend to the ground and the Site Supervisor must report back immediately to the Project Manager.

Examples of inclement weather conditions:

- Rain
- Lightning
- Extremely cold weather
- Extremely hot weather
- Sleet or snow
- High or gusty winds
- Poor visibility (fog or smog)



- **Additional Hazardous Environmental Conditions:**

Bee awareness

Honey Bees are an endangered species, but is also a hazard in the work-at-height industry.

They are responsible for pollinating 70 of the 100 crop species that feed 90 % of the word and this is why we should not kill them, but rather protect them.



The Do's and Don'ts of bees.

First you need to know when Bees are most active "Prime Time" for the swarms are during spring and summer time.

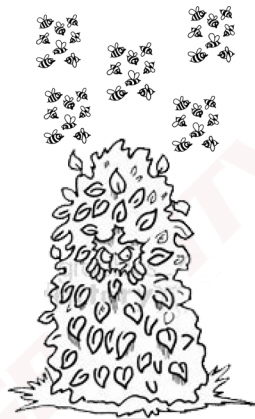
When finding a bee colony unexpectedly:

- Do not approach closer.
- Do not make sudden movements.
- Slowly but efficiently move away.
- Bees have a way of warning you in different ways, by fly/bump into you or they will fly more actively/ making more noise.
- Listen and adhere to the warnings.
- Do not swat at bees, this will only cause them to be aggravated.
- Stay calm. Bees can feel if you are stressed and then they will get stressed.



If you are attacked by bees:

- Do not stand still and swat at them
- Run away in a straight line and do not zig-zag, you will cover a bigger area of ground running in a straight line.
- Keep running for at least a KM or to the closest enclosed space.
- If not in an urban area run into a dense brush, pull the thickest brush around your head. A few will sting but most bees will relent and move away.
- Do not scream and panic.
- Bees will enter your mouth and sting you.
- Try and keep your head covered.
- If you manage to take shelter with a locked door, switch off the lights. Bees are attracted to light.
- Do not jump into water. The bees will wait until you come up to breathe.



DO NOT:

- Throw things at bees.
- Try to burn them out.
- Try to smoke them out.
- Try to poison or kill bees.
- Attempt to move the bees yourself.

**It is important to:**

- Always observe your surroundings.
- Remember that bees are everywhere.
- If you do get stung, remove the sting by scraping the surface of the sting with a Card or something similar.
- Do not pull out the stinger, by pulling/ pinching on it you may press the venom sack.
- Call a professional to move the bee colony.
- Remember that Smoking bees is a delicate process.
- If you are allergic ensure to have an “EpiPen” to prevent you from going into Anaphylactic Shock.

When Visiting a site:

- Check for any quick movement of flying, this is characteristic of the honey bee.
- If observed stop and look for any activity, the activity will give an indication of where the colony made a nest.

After identifying the nest, the following needs to be done:

- Take photos of where the bees have made their nest.
- Inform your management about the nest in order for him to get a professional to move the bees.



- **Additional Hazardous Environmental Conditions:**

Bird awareness

Crows

- Crows are found on every continent except Antarctica, and are among the most intelligent animals on the planet.
- They are able to recognize a single persons face among many other people.
- They are able to even communicate to other crows and share information whether it is good or bad.
- Crows attack humans because they enter their habitat, during breeding season crows defend their eggs or young ones to humans that threaten them.
- If you see a nest or you see a crow on the structure do not advance they will make a noise to warn you before they attack.
- Contact a professional to come and take care of the animal.



Hawks and Eagles

- Hawks and Eagles are also among the most intelligent birds on the planet, but are also a danger people working at height.
- With most birds of prey you stand a chance of being attacked due to specific reasons.
- The most common reason is, it is breeding season and you are close to their nest posing a danger to their eggs or young ones.
- Hawks and Eagles will warn you by making noise before they attack.
- By entering their habitat/lifestyle you automatically pose a threat to them because they think you want to injure them and take their food and or water, so it is within their nature to protect themselves.
- Do not attempt to deal with them by yourself, get a professional to come take care of the animal.
- Inform your manager about them and take photos as prove.



STEP 4 : POST-MITIGATION RISK RATING

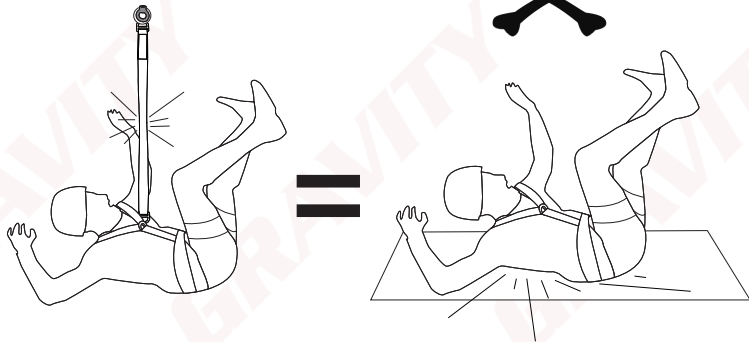
		HAZARD SEVERITY				
		Negligible	Slight	Moderate	High	Very High
LIKELIHOOD OF OCCURRENCE	Very Unlikely	LOW	LOW	LOW	LOW	LOW
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B. Work at Height Basics

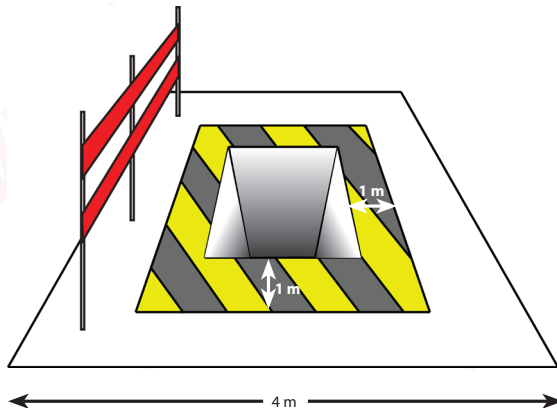
1. Shock Load

The force felt by a body when a fall is arrested within a short amount of distance and a short duration of time



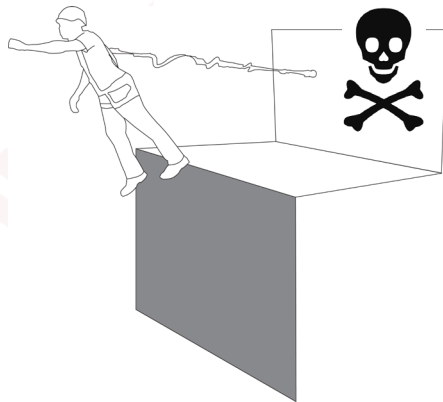
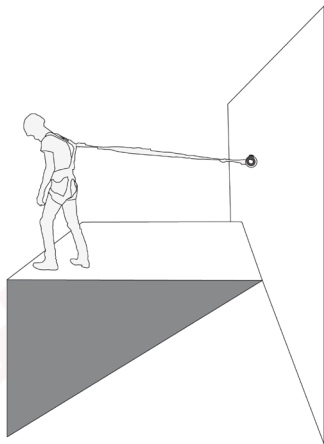
2. Exclusion Zones and Barricading

- **Exclusion zones:** A zone or space where people are **not allowed** to walk or work within, for example, a 1 m space around a hole in a roof.
- **Barricading:** A physical barrier **preventing** people from reaching a fall risk position.



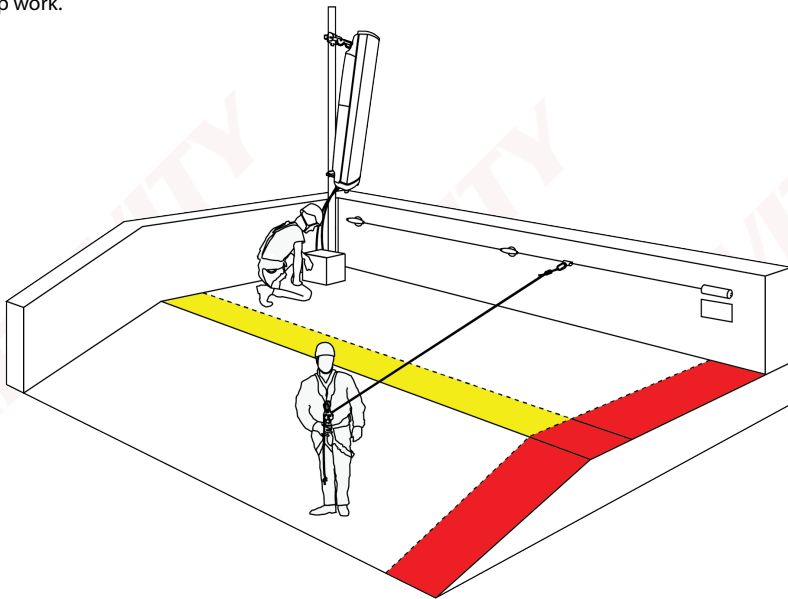
3. Work Restraint

A system where we physically tie a worker to the structure, thus preventing him/her from reaching the edge, thus preventing a fall.



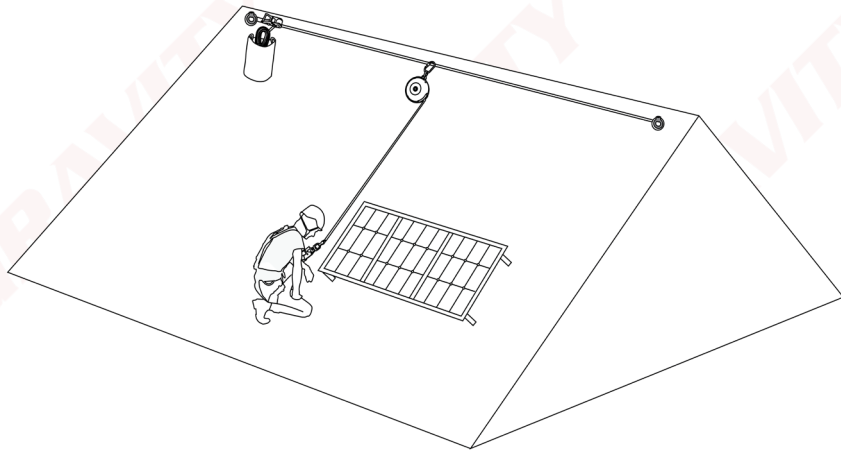
4. Roof Top Work Restraint System

One can combine the horizontal lifeline with a work restraint system to create a system ideal for rooftop work.



4. Pitched Roof Work

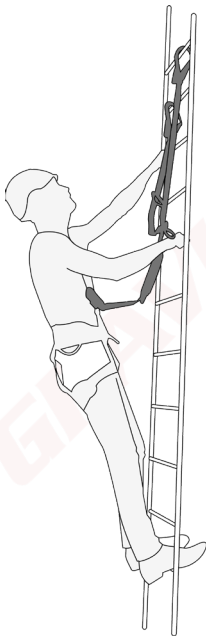
One can combine the temporary horizontal lifeline with a retractable lifeline to create a system ideal for pitched roof work.



5. Fall Arrest

A system used to suspend a person safely AFTER a fall has occurred, and doing all this in such a way as to prevent injury as far as possible.

Fall Arrest does NOT prevent a fall like Work Restraint. It only prevents a person from serious injury or hitting the ground due to a fall.



6. Full Range of Personnel Protective Equipment (PPE)

In order for a user to use PPE safely and correctly, there are four things each user must know about each piece of equipment, namely:

- **Identification** - The name and purpose of each piece of equipment.
- **Certification** - Only equipment with the correct South African National Standards (SANS) or European National Standards (EN) should be used and must be purchased through a reputable retailer such as Gravity Training. For example, Gravity Training only makes use of ISO 9001:2015 certified equipment suppliers.
- **Limitation** - How to use and not use equipment, the safe use criteria and hazards to identify when using each piece of equipment, and the maximum load that each piece of equipment is certified to carry. Equipment load bearing ability is given by the MANUFACTURER as **Safe Working Load** and the **Breaking Strength** and should never be exceeded.
- **Inspection** - Pre-use inspection should be done before start of work and that equipment should be sent to the appointed equipment controller for detailed 3 monthly inspections.
- **Note that the S.W.L will differ from one manufacturer to another.**

Climbing Helmet (EN 12492)



A helmet is used to protect the technician's head during a fall as well as from falling objects:

- Must have chinstrap.
- Must be on worker's head at all times.
- Chinstrap is designed to break under force of 50 kg.
- Is designed in the way that it can take top impacts and side impacts.

Construction Helmet EN 397



- Do not use a EN 397 Helmet for working at height.
- Reasons why one should not use these helmets, they are not rated for side impact forces.
- Should an object fall on the front of the helmet it will lift up the back of the helmet and expose the users head to potential injury.

Full Body Harness (EN361& EN358)



A harness allows all other devices and systems to be connected to the worker. This forms the basis of all safety systems.

- Fall arrest systems must preferably be connected the chest D-link.
- The two buckles on the side of the harness are for work positioning systems.
- The harness must be fitted correctly with all straps tightened properly.
- Ensure that the buckles go to the correct corresponding connectors.
- S.W.L = 140kg
- B.S = 1400kg

Donning and duffing of a harness

When donning a harness ensure the following:

The waist belt is above your hips and tight.

The shoulder straps are tight so that you can fit a fist between your chest and the harness.

The leg loops are adjusted so that they do not restrict blood flow when you crouch.

All the buckles are connected and secure.

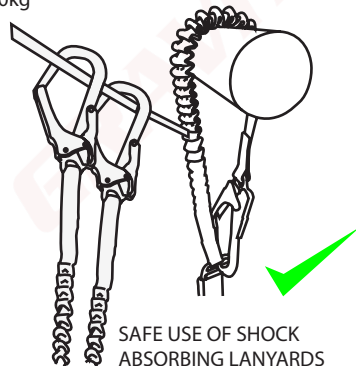
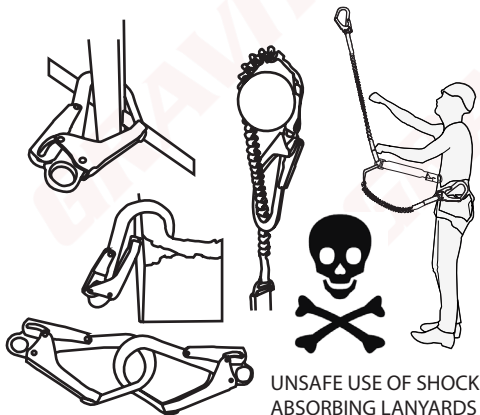


Shock Absorbing Lanyard (EN 355)

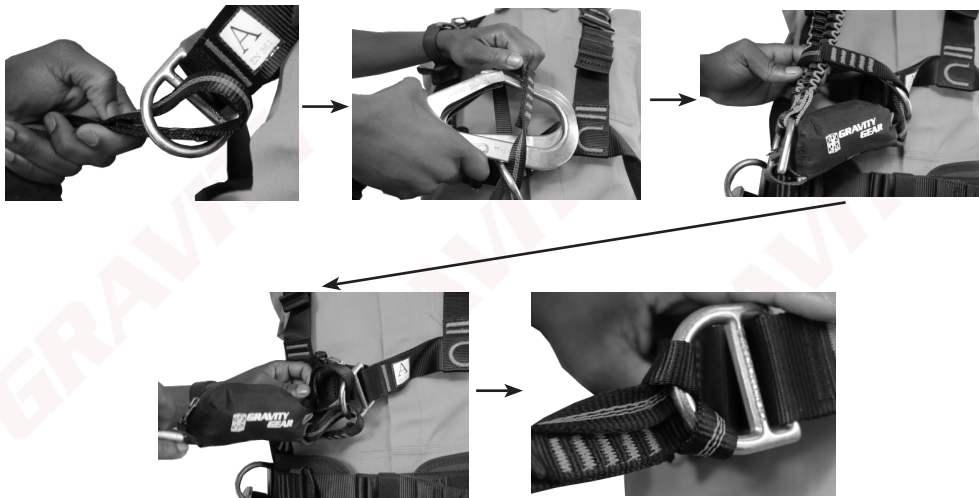


A shock absorber acts as the Primary safety when working at height.

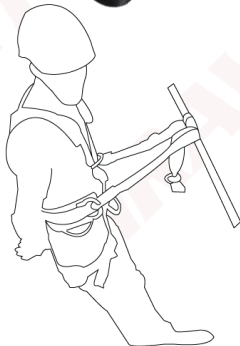
- Shock absorber rips open (deploys) to reduce the shock load felt by the person falling.
- Must be intact during use. Once activated, the shock absorber may not be used again as it will not be able to absorb the shock load of another fall.
- S.W.L. = 130kg
- B.S. = 1300kg



How to fit the Gravity Shock Absorbing Lanyard



Work Positioning Lanyard (EN 358)

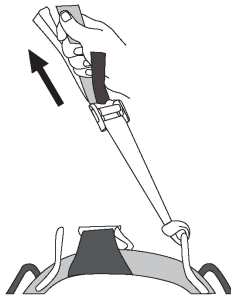
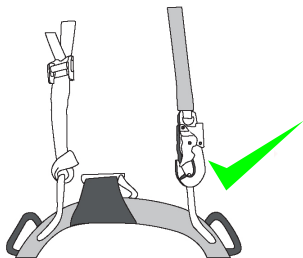


A work positioning lanyard is used to allow a worker to use both hands freely in order to do work effectively.

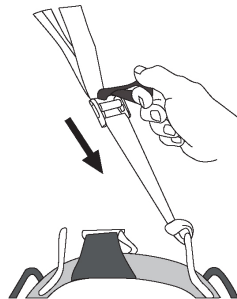
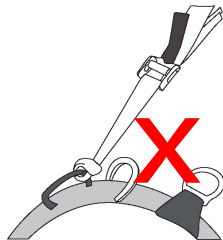
- MUST NEVER BE USED WITHOUT A FALL ARREST SYSTEM.
- Should be wrapped around the structure to prevent slipping.
- S.W.L. = 140kg
- B.S. = 1400kg

The helmet, harness, shock absorber and work positioning lanyard form the minimum PPE that must be used by a worker when at height.

How to fit the Gravity Work Positioning Lanyard



How to adjust the Gravity Work Positioning Lanyard



Rope Grabs Devices (EN 353-2)



Used on a vertical lifeline as a Fall Arrest device. This will grab the rope in the event of a fall, limiting the falling distance to only rope stretch.

- Must be used in a safe fall factor.
- Used on a vertical lifeline.
- Must be used with a mouse tail in order to prevent the user from interfering with the working action of the device.
- S.W.L = 120kg
- B.S = 1200kg



Descending Devices (EN 341)



A variable friction device that controls the rate of descent. Used for an emergency escape, horizontal lifelines and rescue.

- Must be locked when not in use.
- Used during emergency escape, when tensioning a horizontal lifeline and during rescue.
- When using in rescue, extra friction must be applied.
- S.W.L = 120kg
- B.S = 1200kg
- S.W.L (with extra friction) = 220kg



Slings (EN 566 & 795)

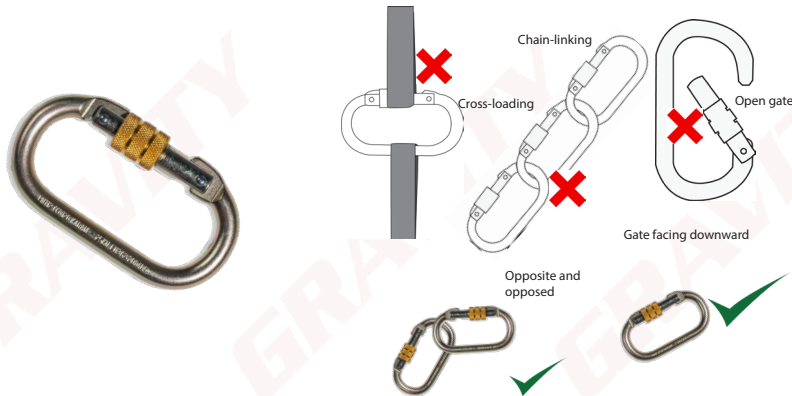


Used to anchor systems to structures when sharp edges are not present.

- Should be wrapped around the structure to prevent slipping.
- Stitching must be straight to prevent the weakening of the sling.
- Slings with PVC sleeves should be used should sharp edges be a concern.
- S.W.L = 220kg
- B.S = 2200kg

Connectors / Karrabiners (EN 362)

Used to connect a range of devices and different pieces of equipment.



- S.W.L = 220kg
- B.S = 2200kg

Work Ropes

Semi-Static Ropes (EN 1891)

Ribbon stating date of manufacture

Kern\Core

Sheath\Mantle



Used throughout work at height systems and is a crucial part of height safety.

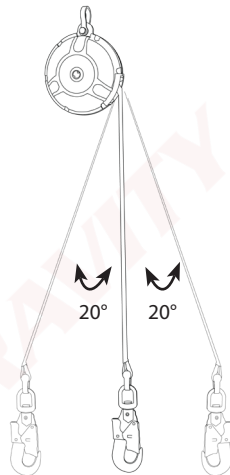
- Stretches between 5% and 10%.
- Is of kermantle construction. Consists of two parts:
 - Kern (Core) = 80% of the strength.
 - Mantle (sheath) = 20% of the strength.
- S.W.L = 250kg
- B.S = 2500kg

Retractable Lanyard (EN 360)

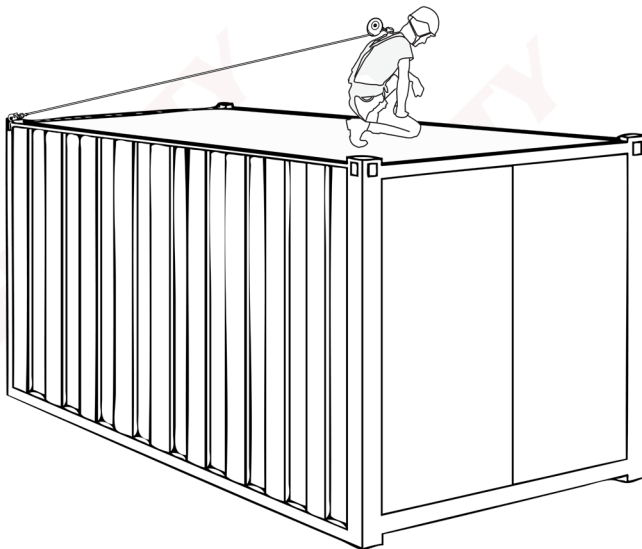


Used as a self-retracting fall arrest device, this allows quick and easy movement up and down. The device limits falling distance to approximately 30 cm.

- Similar functionality of a car safety belt; it allows a technician to freely move up and down, and locks only when a person has fallen.
- Can be used for work positioning and fall arrest.
- Some models include a shock-absorbing function.
- Has a 20 degree maximum range during vertical use.
- Never use above a Fall Factor 1.
- S.W.L. = 120kg
- Please note the S.W.L might differ from manufacturer or by design.



Using Retractable Lanyard while working on containers



- When using it on a rooftop or container the device should connect to the technician and not to the structure, to avoid any reduction in the stopping distance. However not all Retractable are rated for this work.

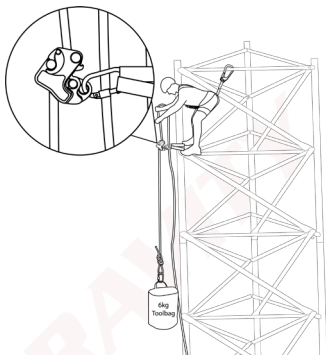
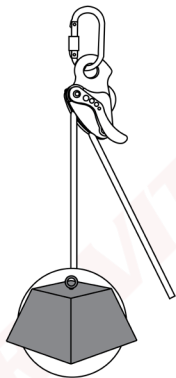
Carrying Tools at Height

All tools must be tied off with tool lanyards or accessory cord to prevent tools from being dropped from height. Tools being dropped can cause damage to property and even injury or death to persons below.



Tools weighing 8 kg or less should be connected to the harness gear loops. Tools weighing in excess of 8 kg should be lifted using a separate 1:1 pulley system with a maximum lifting capacity of 20 kg. Any tools or equipment weighing more than 20 kg need to be lifted by a qualified rope rigger. A qualified rope rigger may lift up to 40 kg with using a 1:1 system and up to 100 kg using a 3:1 system.

Direct Lifting



Principles of Slings:

- Never exceed the safe working load of equipment.
- Always protect slings from sharp edges.
- Always perform a pre-lift before lifting.
- Always consider the center of gravity.

Principles of Rigging:

- Never stand underneath a load.
- Always ensure that the route traveled is clear of obstructions.
- Ensure that the correct equipment is used for the lift.

Pre-use Inspections

Users must inspect all the equipment to be used before work starts. This must be done to ensure that all equipment is safe for use and fit for purpose in accordance with ISO 22846-2.

Equipment is divided into 2 categories namely hardware and software, each of these have their own inspection criteria. Some pieces of equipment (e.g a harness) fall into both categories and must be inspected as such.

Software



Check for:

- Date of manufacture
- Condition of stitching
- Discoloration
- Cuts
- Burns
- Wear and tear
- Certification

Hardware



Check for:

- Rust
- Working action
- Movement
- Wear and tear
- Deformation
- Cracks

Equipment Storage

Equipment must be stored correctly and according to manufacturer's specifications to prevent damage to equipment when not in use.

Store Equipment:

- In a cool, dry place - This will help prevent mildew in software and rust in hardware.
- Free of chemicals - This will reduce the chance of chemical damage to equipment.
- Where there are no infestations (rats and moths etc.), Which can damage software; ensure that there is a system in place to prevent this.
- In a secure location - This will prevent unauthorized personnel entering and potentially damaging equipment.

Rope Care

- Do not use rope for anything other than work at height.
- Do not run rope over sharp edges; use edge protection.
- Do not stand on or walk over a rope.
- Ensure that all devices are compatible with any devices to be used.



Rope Bagging:

It may be preferable to bag a rope instead of coiling it. In such cases, it is recommended to deviate the rope over the structure or ladder and pull the rope into the bag, this will prevent knotting and allows a rope to be bagged faster.



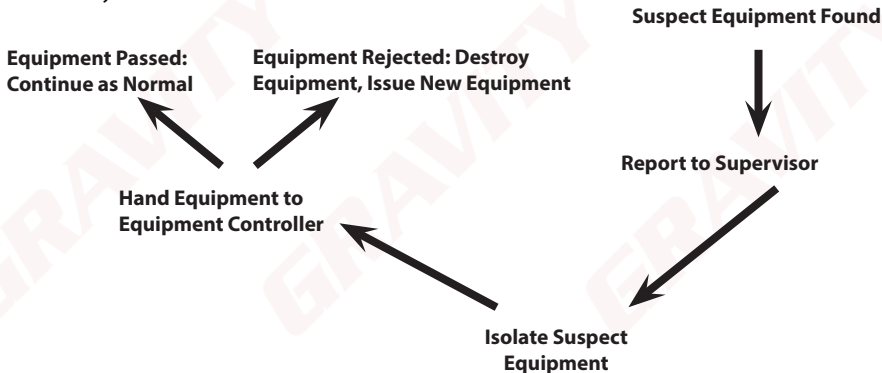
Housekeeping:

Whilst on site, it is important to ensure that the equipment being used is handled in such a way that they do not suffer any unnecessary damage or exposure to environmental conditions. When equipment is not in use, store them in their bags or bins. Ensure that equipment is moved out of the areas where workers move to prevent slips and trips or other accidents.

Reporting Suspect Equipment

Gear that is found to be damaged or too old for use must be reported to the person responsible for the site (either the Site Supervisor or Safety Officer etc).

Damaged or old equipment must be properly marked and removed from service immediately. Damaged equipment must be destroyed at the earliest opportunity to prevent the equipment from getting back into the work system.



Repairing Damaged Equipment

Equipment may only be repaired by either the manufacturer or a company certified by the manufacturer.

NO CONTRACTOR MAY REPAIR PPE.

C. Work at Height Safety Principles

1. Fall Factors

Fall Factor =

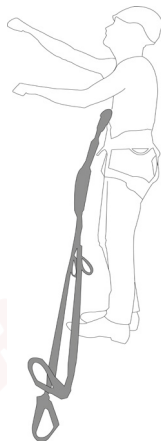
$$\frac{\text{Distance of fall}}{\text{Length of lanyard}}$$



Fall Factor 0:



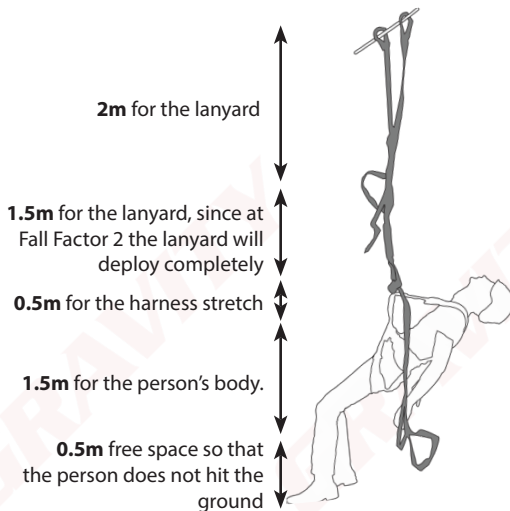
Fall Factor 1:



Fall Factor 2:

1. Minimum Free Space

Minimum free space (MFS) is the open space a technician would need below him to safely arrest a person's fall without hitting the ground or a structure. Reducing the fall distance or Fall Factor would reduce the required MFS.

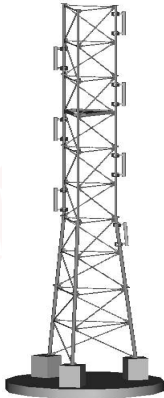


THUS: $2\text{ m} + 1.5\text{ m} + 0.5\text{ m} + 1.5\text{ m} + 0.5\text{ m} = 6\text{ m}$

2. Anchor Points and Lifelines

Anchor points are mainly divided into two categories:

- Self-identified anchor points, and
- Planned anchor points certified to EN 795



Self-identified

SWL = 150 kg



Gravity Access Anchor



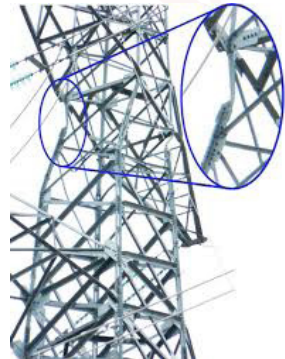
Eye-nut

Planned

SWL = 120 kg

3. Structural Evaluations

- "It is the responsibility of the owner of the structure to ensure the structure is safe to climb and to perform any rigging and lifting activities when needed.
- All structures must be visually inspected before any climbing or lifting operations are conducted. It is not the responsibility of the climber or the rigger to certify the structure, but must only ensure that the structure is safe to climb. Each climber must check the following.
 - That the structure is signed off by an engineer.
 - That the fixed fall arrest system is certified and safe to use.
 - That there is no missing members or loose telecoms equipment.
 - Any roof structure must also be inspected that the surface areas are secure and safe to access."



3. Temporary Vertical Lifelines

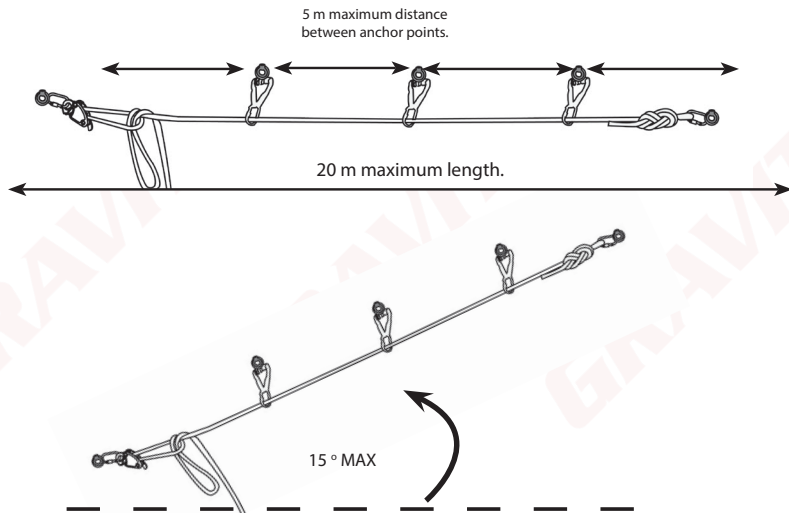
Vertical lifelines are used for easy and quick movement up and down a structure. These lifelines are used with a rope grab device which results in a reduced falling distance, only being limited to the amount of rope stretch.

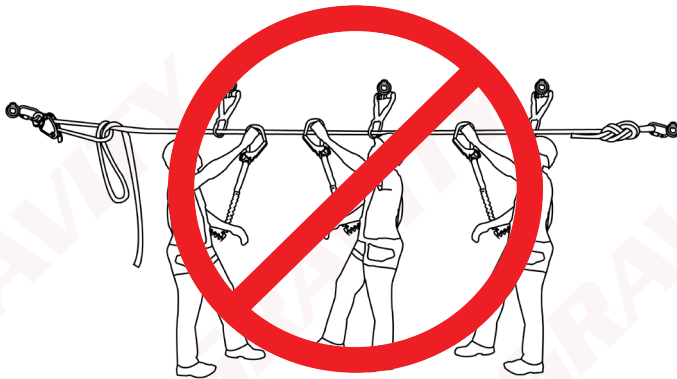
- Maximum Length: 30 M.
- Maximum persons allowed on the lifeline: 1 Person.



4. Temporary Horizontal Lifelines

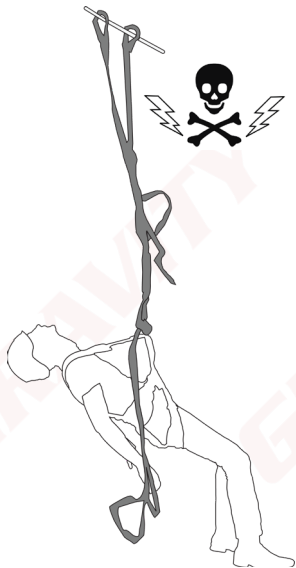
Horizontal lifelines are used for safe and quick movement from side to side. These lifelines are to be used with a shock absorber which can slide from side to side as the technician moves.





- No more than 2 users on a line
- No more than one user per section

2. Suspension Trauma



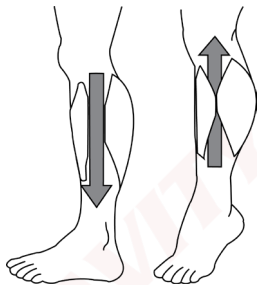
The two biggest risks associated with awaiting a rescue while suspended in a harness, are injury or death due to suspension trauma.

Due to the force of gravity, the blood in the fallen person's body is pulled toward the ground, and pools in the legs.

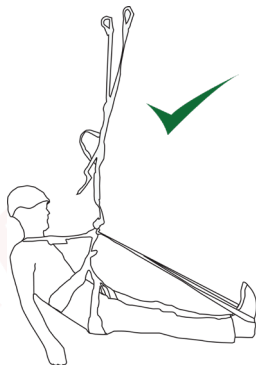
Due to the constriction of the leg loops of the harness, the blood cannot return to the fallen person's heart. This prevents blood being circulated through a person's body. Thus, less and less blood reaches the brain, leading to loss of consciousness and can lead to brain damage. Finally, this can lead to cardiac arrest and eventually death.

All of this can happen within 15-20 minutes; so it is important to have a good knowledge of the techniques needed to prevent suspension trauma, for yourself, and the casualty that needs to be rescued.

Preventing Suspension Trauma



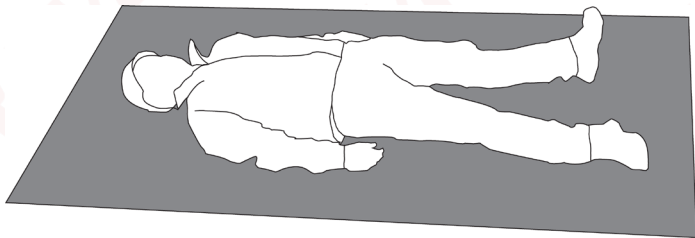
Muscle Pump: Moving your legs, feet and toes will contract the muscles in the legs and force blood back into circulation, which provides the brain with the much needed oxygen to stay conscious.



Suspension Trauma Sling: Will allow you to stand up or to get horizontal, prevent the blood pooling in the legs as well as relieve the pressure of the leg loops which will allow the blood to remain in circulation to the heart and brain. Should the casualty be unconscious, the rescuer can do this for the casualty.

3. Recovery Position

This position allows the blood to flow slowly enough as to prevent Re-flow Syndrome, while at the same time keeping the casualty stable and still as to prevent any possible further injury to the casualty's neck and/or spine. This position is also known as the "Supine Position".

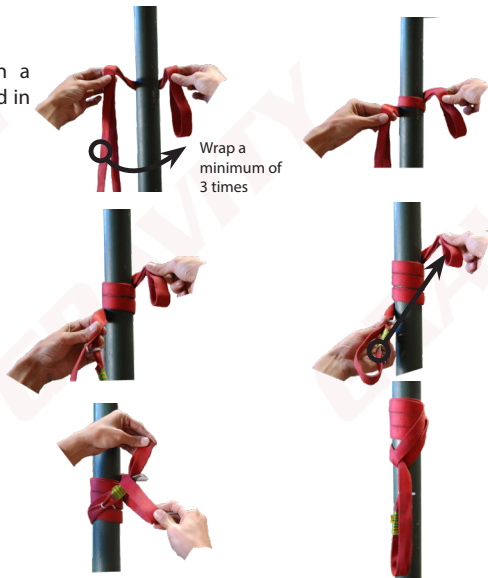


E. Knots

- All knots weaken and decrease the length of the rope.
- All knots must be dressed properly; an incorrectly dressed knot can be up to 50% weaker than a knot dressed correctly.

4. Klemheist

Used to create an anchor point on a vertical or diagonal pole. Must be used in a Fall Factor 1 or better.



The Way Forward

Congratulations on completing Gravity Training's Basic Fall Arrest course. At Gravity we care about the professional development of our clients, should you wish to take the next step you might be interested in enrolling in some of our other courses:

US 229995 - Fall Arrest and Basic Rescue

US 229994 - Fall Protection Plan Developer
(We recommend doing 229995 first)